

Artificial Intelligence

I. Fill in the blanks

1. The term Artificial Intelligence was first coined by John McCarthy at the Dartmouth conference in 1956.
2. An Environment is called Partially Observable .when the agent does not have complete information about its current state.
3. Depth First Search (DFS) search goes deep into the tree before exploring other branches.
4. K-Means algorithm groups similar data points into clusters.
5. Deep learning uses Neural networks .
6. Deep Learning is part of Machine Learning.
7. PEAS stands for Performance measure, Environment, Actuators, Sensors
8. Regression predicts Continuous Values .
9. DFS Uses Stack Structure .

II. Match the following

A

1. Simple Reflex Agent
2. Goal-based Agent
3. Breadth First Search
4. Depth First Search
5. Activation Function
6. Accuracy
7. CV (Computer Vision)
8. Machine Learning

B

Uses condition-action rule
Acts to achieve goal
Explores level wise
Goes deep first
Decides neuron output
Evaluation
Image
Learning from data

III. Multiple Choice Questions

- 1) Which type of AI can perform any intellectual task that a human can do?
a) Narrow AI b) Weak AI **c) General AI** d) Generative AI
- 2) Which of the following is a key limitation in current AI systems?

a) Ability to process large data

b) High accuracy in predictions

c) Lack of common sense reasoning

d) Fast computation

3) In the PEAS description, which component defines the criteria for success of an agent?

a) Environment

b) Performance Measure

c) Sensors

d) Actuators

4) A is:*

a) Informed search

b) Uninformed

c) Blind

d) None

5) DFS uses:

a) Stack

b) Queue

c) Heap

d) None

6) Which algorithm is best suited for predicting continuous values?

a) Logistic Regression

b) K-Means

c) Decision Tree (Classification)

d) Linear Regression

7) Which algorithm can be used for both classification and regression tasks?

a) K-Means

b) PCA

c) Decision Tree

d) A*

8) Deep learning refers to:

a) Learning using multiple hidden layers.

b) Learning with shallow networks.

c) Learning with a single neuron.

d) Learning without data.

9) Which type of neural network is mainly used for image processing tasks?

a) Recurrent Neural Network (RNN)

b) Convolutional Neural Network (CNN)

c) Feedforward Neural Network

d) Radial Basis Network

10. Which pre-processing technique removes common words like "I", "he", and "and"?

a) Tokenization

b) Stemming

c) Stop-word removal

d) Lemmatization

11. PEAS includes

a) Performance

b) Environment

c) Actuators

d) All of the above

12 . Actions are performed by

a) Actuators

b) Sensors

c) Memory

d) None

13. K-means is

a) **Clustering**

b) Classification

c) Regression

d) None

14. AI is used in

a) Healthcare

b) Agriculture

c) Education

d) All of the above

IV. Answer the following

UNIT -1

1. Explain the types of AI with Suitable Examples ?

Types of AI :

Artificial Intelligence can be broadly classified into three main types based on its capabilities:

- 1. Narrow AI**
- 2. General AI**
- 3. Super AI**

1. Narrow AI (Weak AI)

Definition: Narrow AI is designed to perform a specific task or a narrow range of tasks. It operates under predefined rules and cannot perform beyond its programmed function.

Key Characteristics:

- Task-specific
- No self-awareness or consciousness
- Cannot think or reason like humans

Examples:

- Voice assistants (Siri, Alexa, Google Assistant)
- Recommendation systems (Netflix, Amazon)
- Image and face recognition

2. General AI (Strong AI)

Definition: General AI refers to an AI system that can **understand, learn, and apply intelligence across a wide range of tasks**, similar to human intelligence.

Key Characteristics:

- Can reason, plan, and solve problems independently
- Learns from experience

Current Status:

- Does not exist yet

Examples

- A machine that can learn any subject like a human
- AI capable of logical reasoning, emotional understanding, and creativity

3. Super AI

Definition: Super AI is an advanced form of AI that **surpasses human intelligence** in all aspects, including creativity, problem-solving, and emotional intelligence.

Key Characteristics:

- Far more intelligent than humans
- Self-aware and autonomous

Current Status:

- Hypothetical / future concept

Examples (Conceptual):

- Self-improving AI systems
- Highly advanced autonomous robots

2.Explain the real life applications of AI .Also explain the challenges and future scope of AI ?

Real-Life Applications of AI

- **Education:** Personalized learning and smart tutoring systems (e.g., Duolingo).
- **Transportation:** Self-driving cars and traffic management (e.g., Tesla).
- **E-commerce:** Product recommendations and chatbots (e.g., Amazon).
- **Finance:** Fraud detection and automated trading.
- **Smart Assistants:** Voice assistants like Amazon Alexa.
- **Healthcare:** AI helps in disease diagnosis, medical imaging, and drug discovery (e.g., IBM Watson Health).

Challenges of AI

- **Data Privacy & Security:** Risk of misuse of personal data.
- **High Cost:** Expensive development and maintenance.
- **Lack of Transparency:** AI decisions are hard to understand (black-box issue).
- **Job Displacement:** Automation may reduce employment.
- **Bias & Ethical Issues:** AI can produce unfair or harmful outcomes.

Future Scope of AI

- **Advanced Healthcare:** Early diagnosis and robotic surgeries.
- **Autonomous Systems:** Self-driving vehicles and smart cities.
- **Human-AI Collaboration:** AI assisting humans in decision-making.
- **Improved NLP:** More natural interaction with machines.
- **General AI (AGI):** Development of human-like intelligence systems.

3. Define Artificial Intelligence ? Explain the foundations of AI.

Definition of Artificial Intelligence (AI) : Artificial Intelligence (AI) is a branch of computer science that focuses on creating machines or software that can perform tasks which normally require human intelligence.

Foundation of Artificial Intelligence (AI)

1. Computer Science

Computer science is the backbone of AI. It provides algorithms, data structures, programming languages, and software systems required to design and implement intelligent machines. Efficient computation and system architecture are essential for AI applications.

2. Mathematics

Mathematics supports AI through linear algebra, probability, statistics, and calculus. These areas are crucial for data analysis, optimization, machine learning models, and decision-making under uncertainty.

3. Logic and Philosophy

Logic helps AI systems reason, make decisions, and solve problems using formal rules.

Philosophy contributes concepts such as knowledge representation, ethics, and the nature of intelligence.

4. Psychology

Psychology studies human behaviour and learning processes. AI uses these insights to model how humans learn, remember, and respond, helping machines imitate intelligent behaviour.

5. Cognitive Science

Cognitive science explains how humans perceive, think, and solve problems. It helps AI systems replicate perception, memory, reasoning, and language understanding.

6. Neuroscience

Neuroscience studies the human brain and nervous system. It inspires artificial neural networks, which are used in deep learning to recognize patterns and process information.

7. Machine Learning and Data

Machine learning allows AI systems to learn from data rather than relying only on predefined rules. Large datasets help AI improve accuracy and adapt over time.

8. Engineering and Robotics

Engineering applies AI theories to real-world systems. Robotics combines AI with sensors, actuators, and control systems to create intelligent machines that interact with the physical world.

UNIT – 2

1. List and explain the types of environments with examples ?

List of Environments in AI

An **environment** is everything that surrounds an agent and affects its actions.

1. Fully Observable vs Partially Observable
2. Deterministic vs Stochastic
3. Episodic vs Sequential
4. Static vs Dynamic
5. Discrete vs Continuous
6. Single-Agent vs Multi-Agent

1. Fully Observable vs Partially Observable

- **Fully Observable:** Agent can see the complete state of the environment.
Example: Chess game.
- **Partially Observable:** Agent has limited or incomplete information.
Example: Driving a car in traffic.

2. Deterministic vs Stochastic

- **Deterministic:** Next state is completely determined by current state and action.
Example: Solving a mathematical problem.
- **Stochastic:** Outcomes are uncertain or probabilistic.
Example: Weather prediction.

3. Episodic vs Sequential

- **Episodic:** Each action is independent of previous actions.
Example: Image classification.
- **Sequential:** Current decisions affect future outcomes.
Example: Playing chess.

4. Static vs Dynamic

- **Static:** Environment does not change while the agent is deciding.
Example: Crossword puzzle.
- **Dynamic:** Environment changes over time.
Example: Stock market.

5. Discrete vs Continuous

- **Discrete:** Finite number of states and actions.
Example: Chess.
- **Continuous:** Infinite states/actions (real values).
Example: Self-driving car steering.

6. Single-Agent vs Multi-Agent

- **Single-Agent:** Only one agent is involved.
Example: Solving a puzzle.
- **Multi-Agent:** Multiple agents interact.
Example: Multiplayer games or traffic systems.

2. Define Agents and Environments ? Explain Simple reflex agent with neat diagram ?

Definition of Agent - An agent is anything that can perceive its environment through sensors and act upon the environment using actuators.

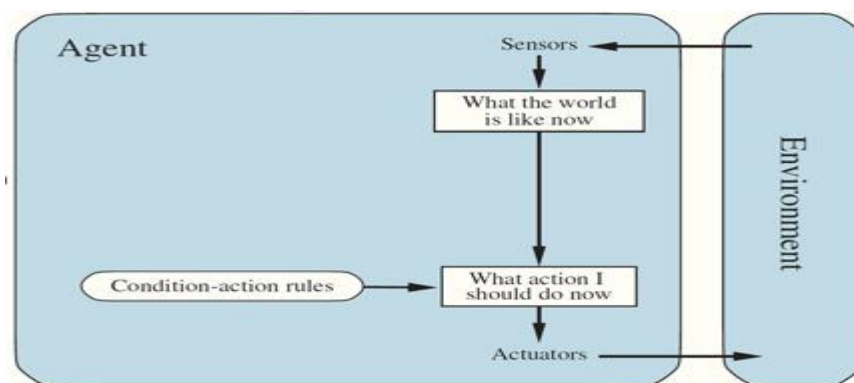
Definition of Environment - An environment is everything that surrounds the agent and affects its actions.

Simple Reflex Agent :

A **Simple Reflex Agent** is an AI agent that makes decisions **only based on the current percept (input)** from the environment, without considering past experiences.

Characteristics :

- Uses only current input
- No memory of past states
- Fast and simple
- Works well in fully observable environments



Simple reflex agent

Working

- The **sensors** receive information from the environment (“what the world is like now”).
- The agent checks this input using **condition-action rules** (if-then rules).
- Based on the rule, it decides “**what action should be done now.**”
- The **actuators** then perform the action on the environment.

Example - Vacuum cleaner agent:

- If dirt is detected → clean
- If no dirt → move forward

3.Explain the PEAS description with an example ?

PEAS stands for:

- **P – Performance Measure**
- **E – Environment**
- **A – Actuators**
- **S – Sensors**

It is a framework used to **define and design an intelligent agent** by specifying what it should do and how it interacts with its environment.

1. Components of PEAS

Performance Measure

- Defines how success of the agent is evaluated.
- It specifies the goals to be achieved.
- *Example:* Accuracy, speed, safety.

Environment

- The surroundings in which the agent operates.
- It includes all external conditions affecting the agent.
- *Example:* Roads, obstacles, users.

Actuators

- The components through which an agent **acts on the environment**.
- *Example:* Wheels, motors, speakers.

Sensors

- The components through which an agent **perceives the environment**.
- *Example:* Cameras, microphones, GPS.

Example of PEAS: Self-Driving Car

Component	Description
Performance Measure	Safety, minimum travel time, fuel efficiency, passenger comfort
Environment	Roads, traffic signals, other vehicles, pedestrians, weather conditions
Actuators	Steering wheel, accelerator, brakes, gear system
Sensors	Cameras, radar, GPS, speed sensors

A self-driving car uses sensors to collect data from the environment, processes it using AI algorithms, and uses actuators to control the vehicle. The goal is to achieve safe and efficient driving based on the performance measures.

Unit -3

1.Explain problem solving as search process. Describe the components of a search problem with an example.

Problem Solving as a Search Process

When an AI agent is given a task (like finding the shortest route on a map or solving a puzzle), it doesn't always know the answer immediately. Instead, it:

- **Formulates a Goal:** Defines what a successful outcome looks like.
- **Formulates a Problem:** Defines the states and actions available.
- **Searches:** Systematically examines different sequences of actions to see which one leads to the goal.
- **Executes:** Once a path (the solution) is found, the agent performs the actions.

Components of a Search Problem - To define a search problem formally, we use five core components:

Component	Description
Initial State	The starting point where the agent begins its search.
Actions	A description of all possible moves available to the agent in any given state.
Transition Model	A description of what each action does (i.e., if I am in State A and take Action X, I will end up in State B).

Component	Description
Goal Test	A condition that determines whether a given state is the solution.
Path Cost	A numeric value assigned to a path. The agent usually tries to find the path with the lowest cost (e.g., shortest distance or time).

Real-Life Example: The 8-Puzzle

Imagine a 3x3 grid with 8 numbered tiles and one empty space. The goal is to slide the tiles until they are in numerical order.

Applying the Components:

- **Initial State:** Any random configuration of the tiles on the board.
- **Actions:** Moving the "blank space" in one of four directions: **Left, Right, Up, or Down**.
- **Transition Model:** If the blank is at the center and we move it "Up," the tile above it swaps places with the blank.
- **Goal Test:** Does the current board match the target state (e.g., tiles 1 through 8 in order)?
- **Path Cost:** Each move typically costs 1. The total cost is the number of moves taken to reach the goal.

2.What is informed search ?Explain A* algorithm in detail ?

Informed search uses a **heuristic function $h(n)$** to estimate how close a current state is to the goal and guides the search in the right direction.

The A* Algorithm

A (*pronounced A-star*)* is the most widely used informed search algorithm. It is popular because it is **complete** (finds a solution if one exists) and **optimal** (finds the path with the lowest cost).

The Evaluation Function

A* selects the next node to expand based on an evaluation function $f(n)$, which combines the cost already spent and the cost estimated to remain:

$$f(n) = g(n) + h(n)$$

- **$g(n)$:** The actual cost incurred to reach node n from the start node.
- **$h(n)$:** The heuristic value, representing the estimated cost from node n to the goal.

- **$f(n)$:** The total estimated cost of the lowest-cost solution passing through node n .

2. Working Mechanism

A* maintains two lists to manage the search process:

1. **Open List:** Contains nodes that have been discovered but not yet fully explored (prioritized by the lowest $f(n)$).
2. **Closed List:** Contains nodes that have already been evaluated.

Steps:

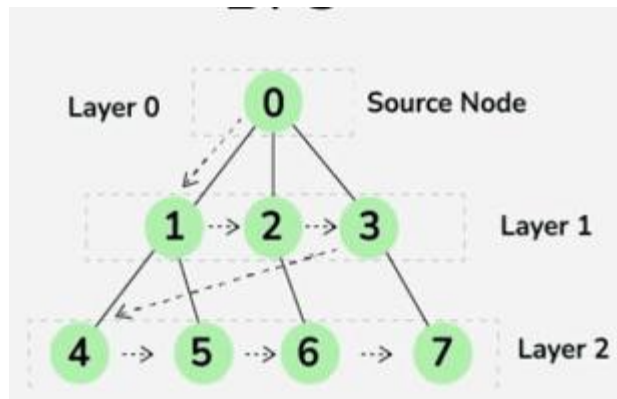
- **Step 1:** Put the starting node in the Open List.
- **Step 2:** Pick the node with the lowest $f(n)$ from the Open List.
- **Step 3:** If the selected node is the goal, the search stops.
- **Step 4:** If not, move the node to the Closed List and expand its neighbors.
- **Step 5:** For each neighbor, calculate $f(n)$. If the neighbor is already in the Open List with a higher cost, update it with the new, lower cost.
- **Step 6:** Repeat until the goal is reached.

3. Explain the Breadth first search(BSF) algorithm with example ?

Breadth First Search (BFS) is an uninformed search algorithm that explores the search space level by level, visiting all nodes at the current depth before moving to the next level.

Algorithm Steps

1. Start with the initial node
2. Insert it into a queue (FIFO)
3. Remove the front node from the queue
4. Check if it is the goal state
5. If not, expand the node and add all its child nodes to the queue
6. Repeat the process until the goal is found



BFS Traversal

Start from the **source node: 0**

Level-wise traversal:

- **Level 0:** 0
- **Level 1:** 1, 2, 3
- **Level 2:** 4, 5, 6, 7

Step-by-Step BFS Execution (Queue Method)

1. Start with node **0** → Queue = [0]
2. Visit **0**, add its neighbors → Queue = [1, 2, 3]
3. Visit **1**, add neighbors → Queue = [2, 3, 4, 5]
4. Visit **2**, add neighbors → Queue = [3, 4, 5, 6]
5. Visit **3**, add neighbors → Queue = [4, 5, 6, 7]
6. Visit **4** → Queue = [5, 6, 7]
7. Visit **5** → Queue = [6, 7]
8. Visit **6** → Queue = [7]
9. Visit **7** → Queue = []

Final BFS Order:

0 → 1 → 2 → 3 → 4 → 5 → 6 → 7

Unit – 5

1 . Define Machine Learning ? Explain the different types of ML with examples ?

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn from data and make decisions or predictions without being explicitly programmed.

Types of Machine Learning

1. Supervised Learning
2. Unsupervised Learning
3. Reinforcement Learning

1. Supervised Learning - Supervised learning uses a **labeled dataset** to train a model. The model learns the relationship between input and output and predicts results for new data.

Applications:

- **Classification:** Predicting categories (e.g., spam / not spam)
- **Regression:** Predicting numerical values (e.g., house prices)

Examples:

- Email spam detection
- Medical diagnosis

2. Unsupervised Learning - Unsupervised learning uses **unlabeled data**. The model identifies patterns and relationships without predefined outputs.

Applications:

- **Clustering:** Grouping similar data (e.g., customer segmentation)
- **Dimensionality Reduction:** Reducing features (e.g., PCA)

Examples:

- Customer segmentation
- Market basket analysis

3. Reinforcement Learning - Reinforcement learning is a method where an **agent learns by interacting with an environment** and receives rewards or penalties for its actions.

Applications:

- Robotics
- Self-driving cars

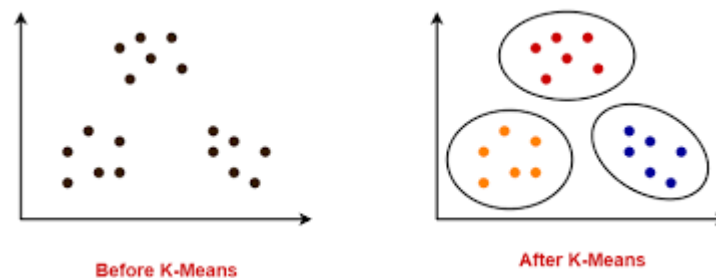
Examples:

- AlphaGo
- Robot learning to walk

2.What is Clustering ?Explain the K – means Clustering algorithm with steps ?

Clustering - Clustering is an unsupervised machine learning technique used to group similar data points into clusters based on their features, without using labeled data.

K- Means Algorithm - K-Means is an unsupervised machine learning algorithm used to divide a dataset into K clusters, where each data point is assigned to the cluster with the nearest centroid (mean value).



Steps of K-Means Algorithm

1. Choose K (number of clusters) - Decide how many clusters you want
2. Initialize Centroids - Randomly select K points as initial centroids
3. Assign Data Points to Clusters - Calculate distance (usually Euclidean distance)
 - Assign each point to the nearest centroid
4. Update Centroids - Compute new centroid as the mean of all points in that cluster
5. Repeat - Repeat steps 3 and 4 until centroids do not change (convergence)

Advantages

- Simple and easy to implement
- Works well for large datasets

Disadvantages

- Need to choose K in advance
- Sensitive to initial centroids
- Not suitable for irregular-shaped clusters

3.Difference between Linear regression and Logistic regression ?

Aspect	Linear Regression	Logistic Regression
Definition	Predicts continuous values	Predicts categorical values (classification)
Output	Numeric value	Probability (0 to 1)
Function	Linear function	Sigmoid (logistic) function
Equation	$y = mx + c$	$\sigma(x) = 1 / (1 + e^{-x})$
Problem Type	Regression	Classification
Output Range	$-\infty$ to $+\infty$	0 to 1
Error Measure	Mean Squared Error (MSE)	Log Loss (Cross-Entropy)
Graph	Straight line	S-shaped curve
Example	House price prediction	Spam email detection

Unit – 6

1.Explain the basic concept of CNN ?Mention their application in real world problems ?

A Convolutional Neural Network (CNN) is a type of deep learning model mainly used for image processing and computer vision tasks.

It automatically learns important features like edges, shapes, and objects from images.

Main Components of CNN

1.Convolution Layer

- Applies filters to input image
- Detects features like edges, textures

2.Activation Function (ReLU)

- Introduces non-linearity
- Removes negative values

3.Pooling Layer

- Reduces image size (down sampling)
- Types: Max pooling, Average pooling

4.Fully Connected Layer

- Final layer for classification
- Converts features into output

Working Process

Input Image → Convolution → ReLU → Pooling → Fully Connected Layer → Output

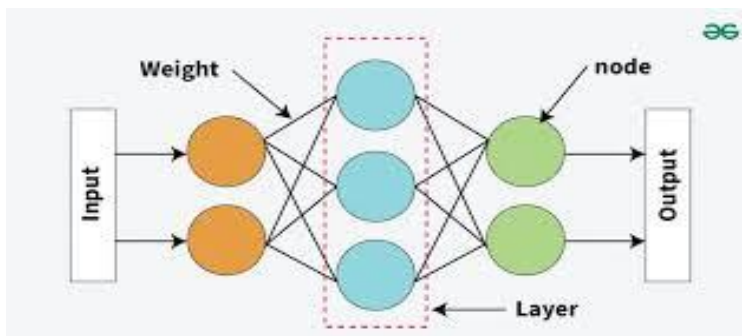
Applications of CNN (Real World)

1. Image Recognition - Face detection, object recognition
2. Medical Imaging - Tumor detection in X-rays, MRI scans
3. Self-Driving Cars - Lane detection, obstacle identification
4. Security Systems - Facial recognition, surveillance
5. Natural Language Processing - Sentence classification

2. Explain the Structure of Neural Network ?

A neural network is a machine learning method inspired by the human brain, using interconnected nodes (neurons) in layered structures to process data, recognize patterns, and make decisions.

Neural Network Architecture



A. Input Layer

- Role: This is the entry point for the data (e.g., pixel values of an image or numerical features of a dataset).
- Neurons: The number of neurons here is exactly equal to the number of features in your input data.
- Processing: No computation happens here; it simply passes the signal forward.

B. Hidden Layers

- Role: This is where the "learning" happens. These layers sit between the input and output.

- Deep Learning: If a network has more than one hidden layer, it is considered a Deep Neural Network.
- Function: They extract increasingly complex patterns. For example, in face recognition, the first hidden layer might detect edges, while the second detects shapes like eyes or noses.

C. Output Layer

- Role: This layer provides the final prediction or classification.
- Neurons: The number of neurons depends on the task:
 - Binary Classification: 1 neuron (e.g., Yes/No).
 - Multi-class Classification: Multiple neurons (e.g., Cat, Dog, or Bird).

Unit – 7

1. Define Natural Language Processing(NLP) ? Describe text preprocessing techniques in NLP ?

Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand, interpret, and generate human language (text or speech).

It combines computer science, linguistics, and machine learning to process human language in a meaningful way.

Text Preprocessing Techniques in NLP - Raw text is often "noisy" and unstructured. Preprocessing is the crucial step of cleaning and transforming this raw data into a format that a machine learning model can process effectively.

1. Tokenization

This is the process of breaking down a large body of text into smaller units called **tokens**.

- **Word Tokenization:** Splitting sentences into individual words.
- **Sentence Tokenization:** Splitting a paragraph into separate sentences.
- *Example:* "AI is evolving" → ["AI", "is", "evolving"]

2. Lowercasing

Text is converted entirely to lowercase to ensure the model treats words like "Apple" and "apple" as the same entity. This reduces the complexity of the vocabulary.

3. Stop Word Removal

Stop words are common words (e.g., "is", "the", "at", "which") that appear frequently but carry little unique information. Removing them allows the model to focus on the "content" words that define the meaning of the text.

4. Stemming and Lemmatization

- **Stemming:** Uses a crude heuristic process to chop off the ends of words (e.g., "running" becomes "run", but "studies" might become "studi").
- **Lemmatization:** Uses a dictionary and morphological analysis to return the word to its actual base form (e.g., "better" becomes "good").

5. Noise Removal - This involves stripping away formatting that doesn't contribute to the meaning of the language, such as:

- HTML tags (e.g., <div>)
- Punctuation and special characters (@, #, \$)
- Extra whitespace or non-ASCII characters

6. Part-of-Speech (POS) Tagging

The process of marking up a word in a text as corresponding to a particular part of speech (noun, verb, adjective, etc.) based on both its definition and its context. This helps the model understand the grammatical structure.

7. Vectorization (Word Embedding)

Machines cannot process strings; they process numbers. Vectorization converts tokens into numerical vectors.

- **Bag of Words (BoW):** Counts the frequency of words.
- **TF-IDF:** Weights words based on how unique they are to a specific document.
- **Word2Vec/GloVe:** Creates dense vectors where words with similar meanings are mathematically close to each other.

2.What is Computer Vision ?Explain basic image processing concepts.

Computer Vision is a branch of Artificial Intelligence that enables computers to interpret, analyze, and understand visual information from images and videos.

Basic Image Processing Concepts

Image processing is a **preprocessing step in Computer Vision** used to enhance and analyze images.

1. Image Representation

- Image is represented as a **matrix of pixels**
- Types:
 - **Grayscale Image** (0–255 intensity values)
 - **Color Image (RGB)**

2. Image Acquisition

- Capturing images using cameras or sensors
- First step in computer vision system

3. Image Enhancement

- Improves quality of image
- Techniques:
 - Brightness adjustment
 - Contrast enhancement
 - Noise reduction

4. Image Filtering

- Removes noise and improves features
- Types:
 - Smoothing (blur)
 - Sharpening

5. Image Transformation

- Changing image size or orientation
- Examples:
 - Scaling
 - Rotation
 - Translation

6. Edge Detection

- Identifies boundaries of objects
- Helps in object recognition

7. Image Segmentation

- Divides image into regions or objects
- Important for analysis

8. Feature Extraction

- Extracts important information such as:
 - Edges
 - Corners
 - Textures

9. Image Compression

- Reduces image size for storage and transmission
- Example: JPEG